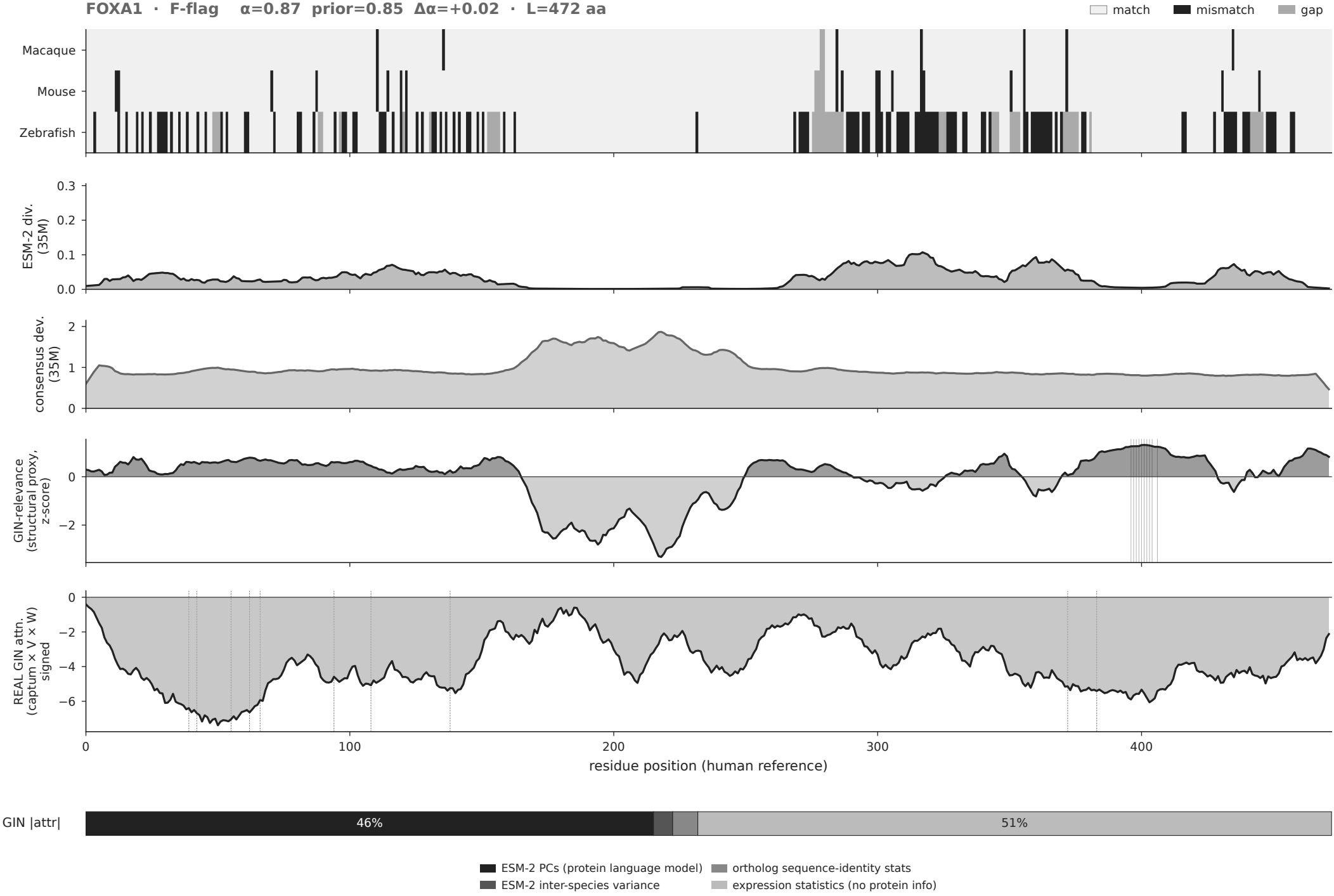

FOXA1 · F-flag α=0.87 prior=0.85 Δα=+0.02 · L=472 aa
match
mismatch
gap
Macaque
Mouse
Zebrafish
ESM-2 div. (35M)
0.3
0.2
0.1
0.0
consensus dev. (35M)
2
1
0
GIN-relevance (structural proxy, z-score)
0
−2
REAL GIN attn. (captum × V × W) signed
0
−2
−4
−6
0
100
200
300
400
residue position (human reference)
GIN |attr|
46%
51%
ESM-2 PCs (protein language model)
ESM-2 inter-species variance
ortholog sequence-identity stats
expression statistics (no protein info)